

EE 122/ME 141 Week 14, Lecture 1: Frequency-domain analysis in presence of noise and disturbances

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1 Announcements

- Problem set #11 due on April 29, 2026
- Project final paper due on May 7, 2026

2 Learning Objectives

By the end of this lecture, you should be able to:

- Understand the effect of noise and disturbances on the performance of a control system in the frequency domain.
- Use the frequency-domain analysis tools, such as Bode plots, to analyze the sensitivity of a control system to noise and disturbances.

The content of this lecture follows directly from the textbook. So, you are encouraged to read the following sections (in order):

- Section 12.1: Sensitivity functions
- Figure 12.1: Explore the block diagram carefully.
- Table 12.1: Derivation of all transfer functions.

- Example 12.1: Need for more than loop transfer function. (you can stop after this example)
- Measuring Specifications on Page 12-12.